

Star Vetter - confirmed eclipsing binary candidates

TIC	Period (d)	Depth	BLS SNR	Pixel SNR	Offset	Tier
437150434	5.2853	15.003%	1081	85.33	1"	Clean
237115819	14.7921	8.883%	507	27.82	4"	Clean
66732476	3.1087	9.814%	370	26.93	8"	Clean
29495157	5.1474	8.579%	333	52.62	5"	Alias
119459681	5.9442	15.977%	311	68.15	2"	Clean
411434067	1.3536	4.289%	193	33.70	4"	Alias
330901906	3.7321	0.498%	60	87.24	4"	Thin margin
303470667	5.4319	2.220%	57	5.02	9"	Marginal
273158289	1.1519	1.255%	41	17.24	2"	Thin margin
111149941	5.2401	0.045%	10	9.73	21"	Ambiguous
251898081	27.3089	18.630%	5535	464.16	3"	Alias
467214801	0.7609	0.950%	2134	21.36	0"	Alias
399770041	1.5802	4.291%	489	36.92	1"	Clean
349701853	30.3427	28.445%	343	51.05	6"	Clean
233459038	25.3384	8.315%	237	84.46	5"	Alias
281310348	1.7702	7.273%	180	57.87	2"	Clean
318887646	14.2817	3.605%	150	6.94	13"	Clean
444465823	18.5133	3.476%	147	25.18	1"	Alias
380943251	0.7209	7.667%	101	15.26	3"	Clean
66716482	8.0283	3.329%	65	21.51	7"	Clean
309656476	28.9204	0.050%	25	13.80	14"	Clean
229494833	15.3839	0.037%	15	5.19	12"	Alias
44533527	14.5952	0.101%	12	7.33	16"	Clean

23 of 151 novel candidates confirmed on-target by TESS pixel-level difference imaging.
Generated by f5_dossier.py from f3i_vet.py, f4f_novelty.py and diag_pixel.py results.
Period column is the BLS-fitted value; candidates tiered Alias have a doubled true period - see their page.

TIC 437150434

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 5.285343 d
Epoch (t0) 2230.8422 BTJD
Duration 0.1700 d
Depth 15.003 % (150,033 ppm)
SNR 1081.0

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

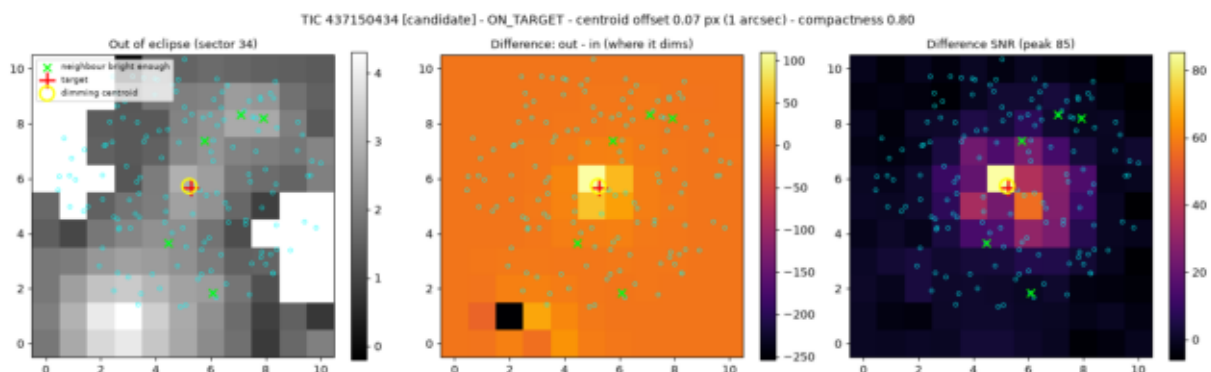
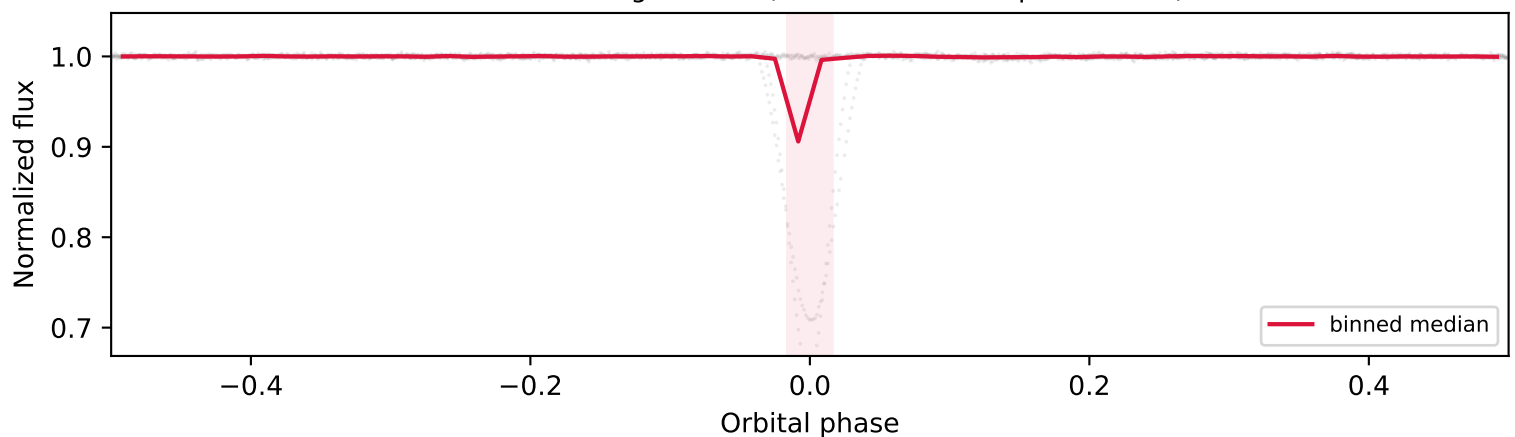
Eclipses observed 4 (gate ≥ 3)
Robust depth ratio 1.18 (gate ≥ 0.50)
Robust depth sigma 891.6 (gate ≥ 7.0)
Odd/even z-score 1.10 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 34 light-curve dip in this sector: 645.4 sigma t0 refined by -0.0075 phase
Difference-image peak SNR 85.33 (gate ≥ 5.0) Compactness 0.80 (gate ≥ 0.50)
Dimming centroid offset 0.07 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 437150402
1.75 px away, dTmag +1.46 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 237115819

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 14.792059 d
Epoch (t0) 1683.4391 BTJD
Duration 0.0840 d
Depth 8.883 % (88,832 ppm)
SNR 507.0

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

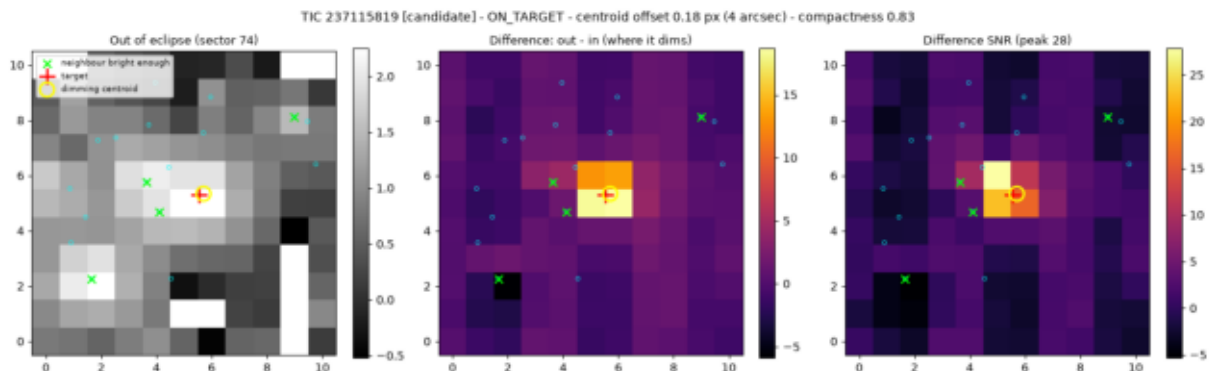
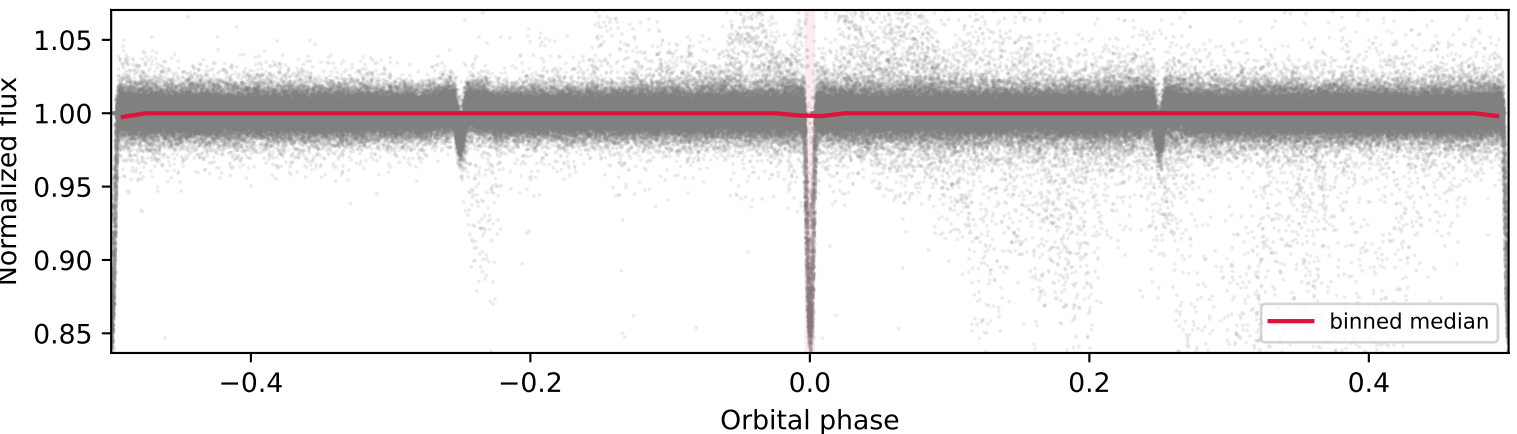
Eclipses observed 75 (gate ≥ 3)
Robust depth ratio 1.23 (gate ≥ 0.50)
Robust depth sigma 414.0 (gate ≥ 7.0)
Odd/even z-score 1.61 (consistent)
Secondary eclipse detected, phase -0.49, 7.9 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 74 light-curve dip in this sector: 51.1 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 27.82 (gate ≥ 5.0) Compactness 0.83 (gate ≥ 0.50)
Dimming centroid offset 0.18 px (4 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 237115821
1.72 px away, dTmag +2.50 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 66732476

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 3.108737 d
Epoch (t0) 2553.3754 BTJD
Duration 0.0840 d
Depth 9.814 % (98,141 ppm)
SNR 369.7

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

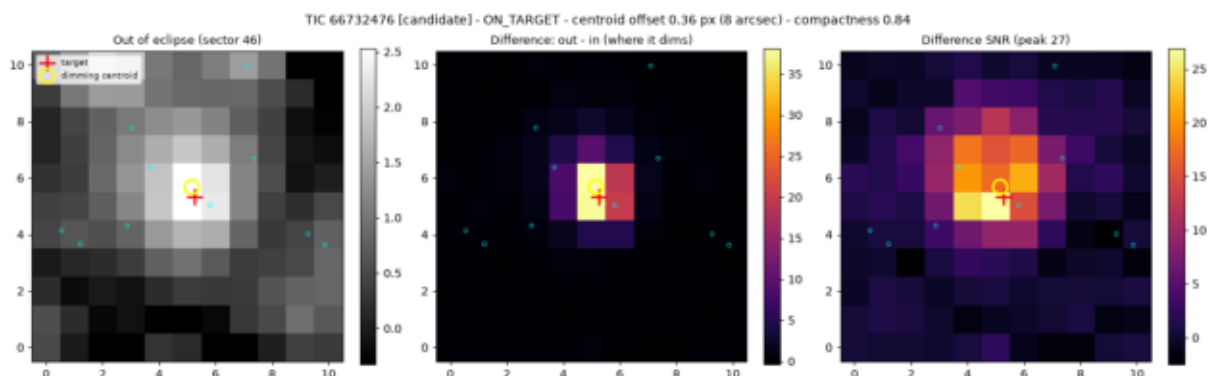
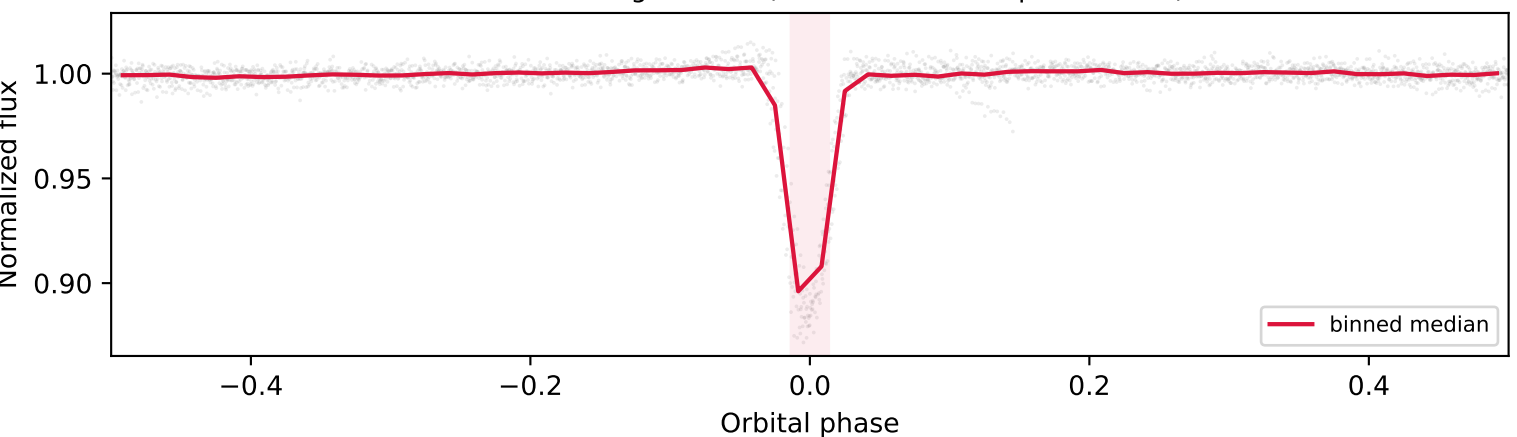
Eclipses observed 8 (gate ≥ 3)
Robust depth ratio 1.06 (gate ≥ 0.50)
Robust depth sigma 247.6 (gate ≥ 7.0)
Odd/even z-score 2.60 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 46 light-curve dip in this sector: 234.7 sigma t0 refined by -0.0050 phase
Difference-image peak SNR 26.93 (gate ≥ 5.0) Compactness 0.84 (gate ≥ 0.50)
Dimming centroid offset 0.36 px (8 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: none within the cutout

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 29495157

PERIOD ALIAS

Eclipsing binary candidate dossier - Star Vetter

BLS EPHEMERIS

Period 5.147449 d (ALIASSED, see below)
-> corrected period: 10.294897 d
Epoch (t0) 1328.1391 BTJD
Duration 0.1700 d
Depth 8.579 % (85,786 ppm)
SNR 333.2

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

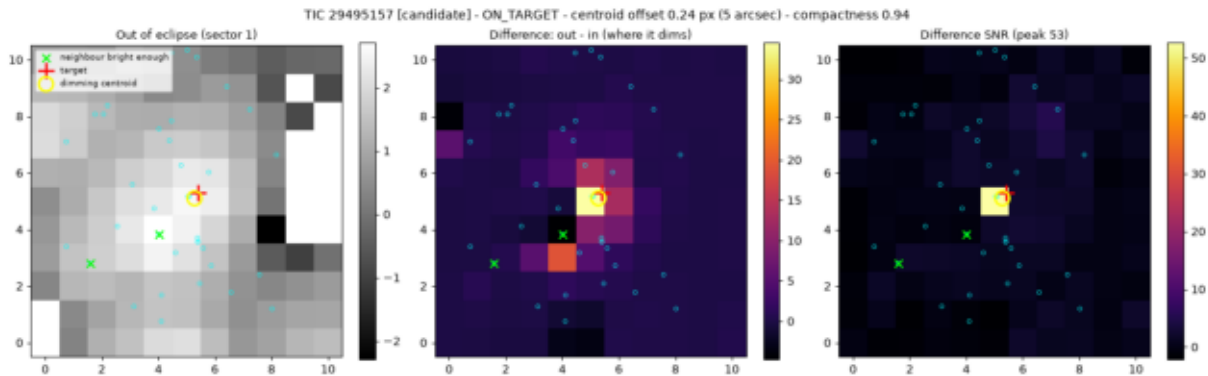
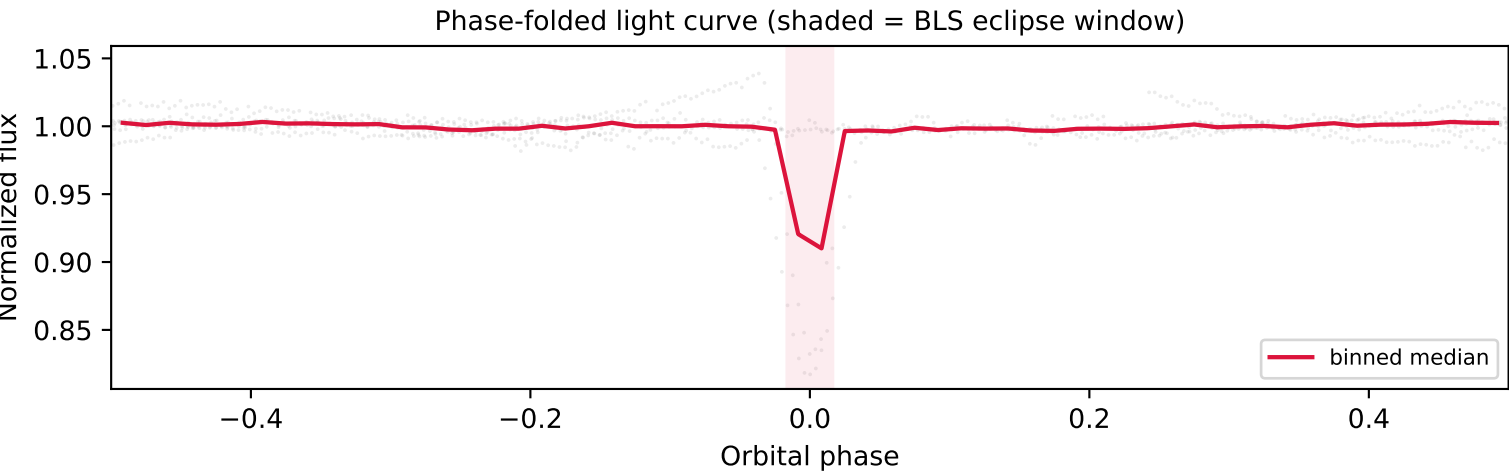
INDEPENDENT CHECKS (f3i gates)

Eclipses observed 4 (gate >= 3)
Robust depth ratio 1.05 (gate >= 0.50)
Robust depth sigma 86.0 (gate >= 7.0)
Odd/even z-score 20.17 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 1 light-curve dip in this sector: 94.0 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 52.62 (gate >= 5.0) Compactness 0.94 (gate >= 0.50)
Dimming centroid offset 0.24 px (5 arcsec) (gate <= 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 29495166
1.79 px away, dTmag -0.48 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 20.17 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 10.294897 d, in any write-up.

TIC 119459681

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 5.944205 d
Epoch (t0) 1471.4023 BTJD
Duration 0.2440 d
Depth 15.977 % (159,771 ppm)
SNR 310.6

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 3 (gate ≥ 3)
Robust depth ratio 1.09 (gate ≥ 0.50)
Robust depth sigma 199.8 (gate ≥ 7.0)
Odd/even z-score 1.19 (consistent)
Secondary eclipse not detected

CATALOG CROSS-MATCH (novelty)

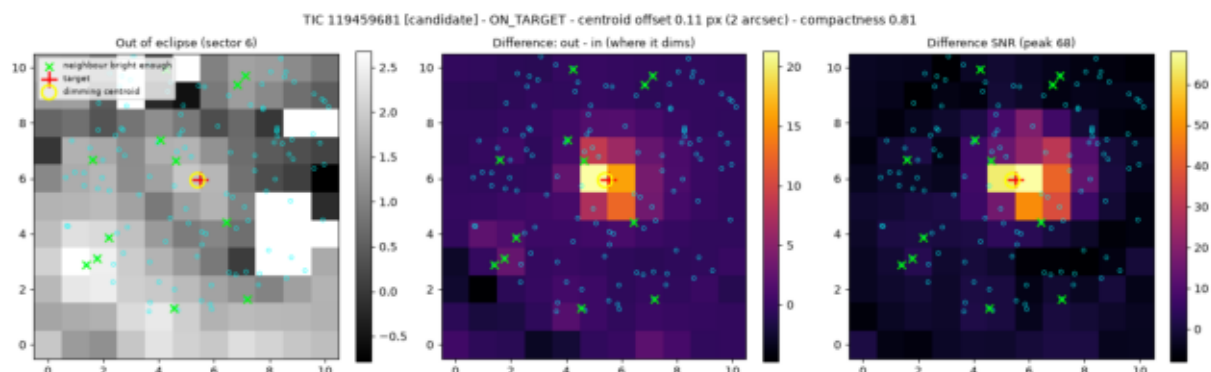
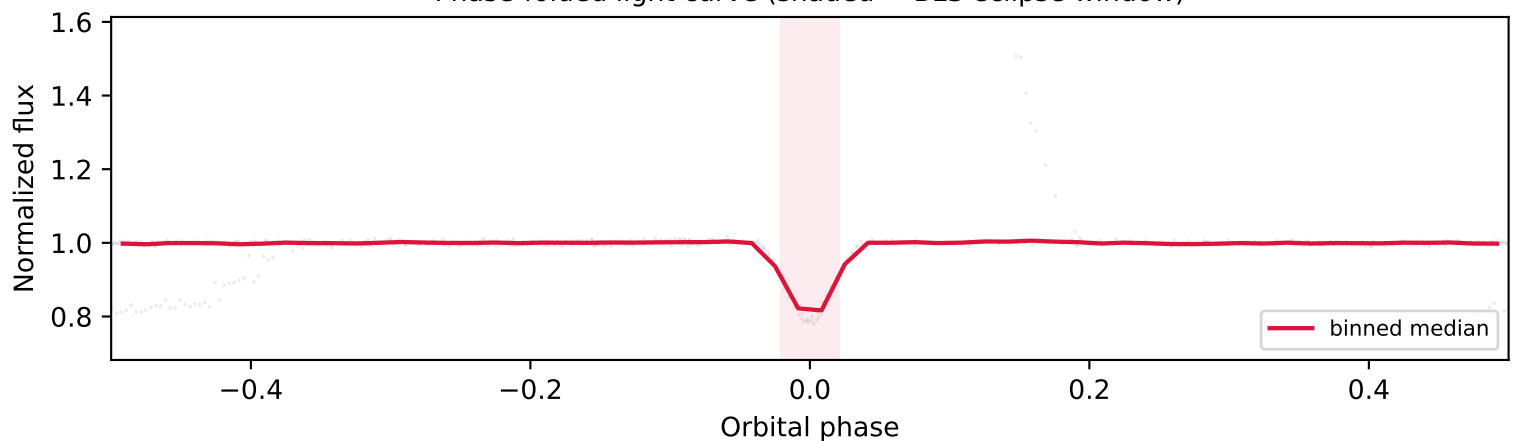
VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 6 light-curve dip in this sector: 183.3 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 68.15 (gate ≥ 5.0) Compactness 0.81 (gate ≥ 0.50)
Dimming centroid offset 0.11 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 119459668
1.05 px away, dTmag +1.18 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 411434067

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 1.353604 d (ALIASED, see below)
-> corrected period: 2.707209 d
Epoch (t0) 2825.8292 BTJD
Duration 0.1200 d
Depth 4.289 % (42,887 ppm)
SNR 192.9

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

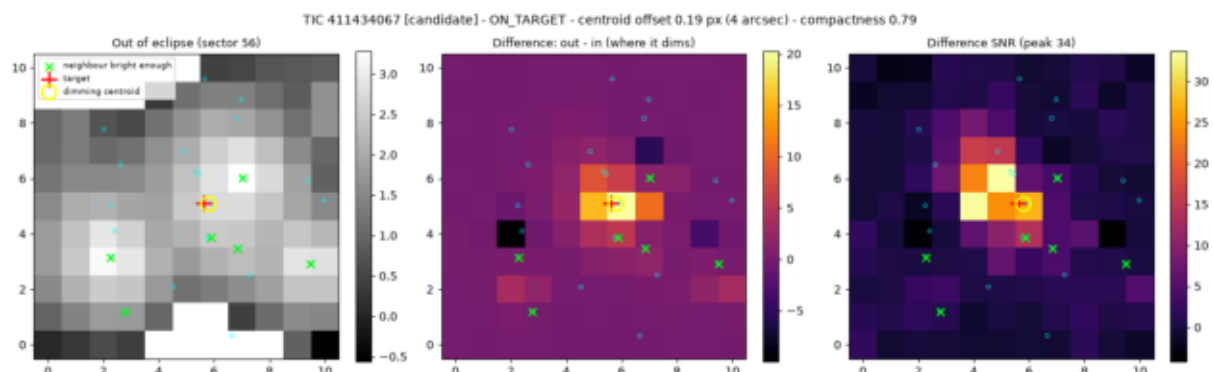
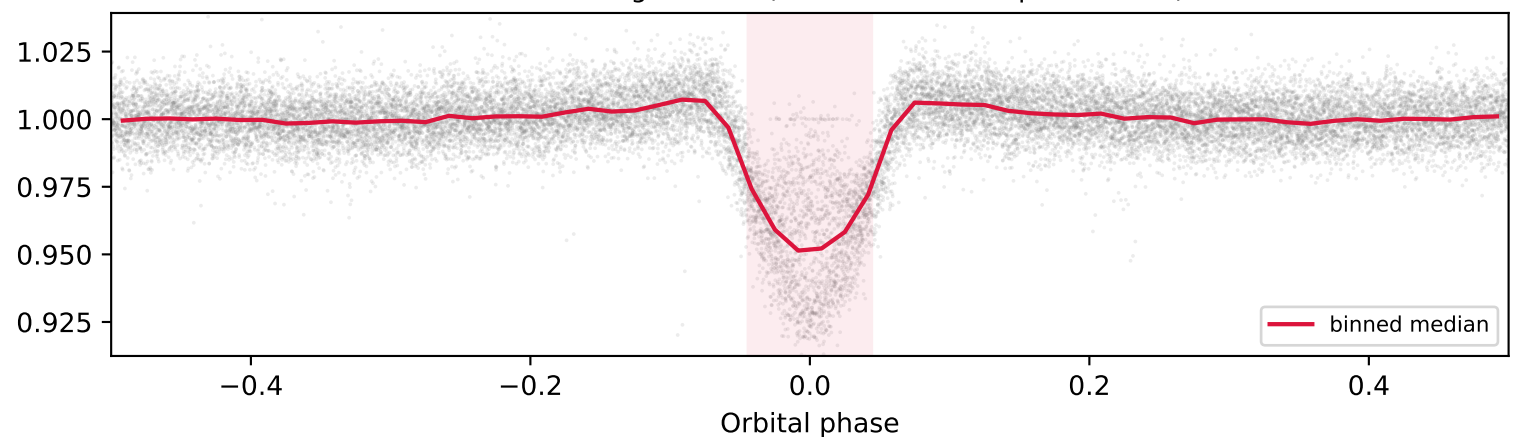
Eclipses observed 39 (gate ≥ 3)
Robust depth ratio 0.92 (gate ≥ 0.50)
Robust depth sigma 162.8 (gate ≥ 7.0)
Odd/even z-score 9.80 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 56 light-curve dip in this sector: 130.8 sigma t0 refined by -0.0025 phase
Difference-image peak SNR 33.70 (gate ≥ 5.0) Compactness 0.79 (gate ≥ 0.50)
Dimming centroid offset 0.19 px (4 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 411429819
1.21 px away, dTmag +1.19 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 9.80 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 2.707209 d, in any write-up.

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	3.732068 d	VSX	no match
Epoch (t0)	1740.9150 BTJD	ASAS-SN	no match
Duration	0.2440 d	Gaia eclipsing-binary table	no match
Depth	0.498 % (4,983 ppm)	Gaia variable classification	no match
SNR	60.1	TESS EB catalog (Prsa+2022)	no match
INDEPENDENT CHECKS (f3i gates)		ExoFOP TOI list	no match
Eclipses observed	29 (gate >= 3)	Verdict:	NOVEL
Robust depth ratio	1.19 (gate >= 0.50)		
Robust depth sigma	58.6 (gate >= 7.0)		
Odd/even z-score	0.36 (consistent)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

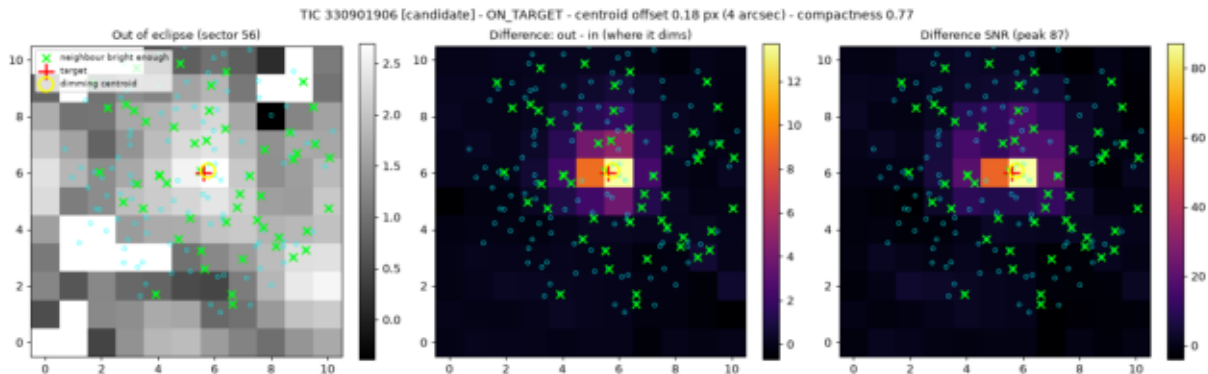
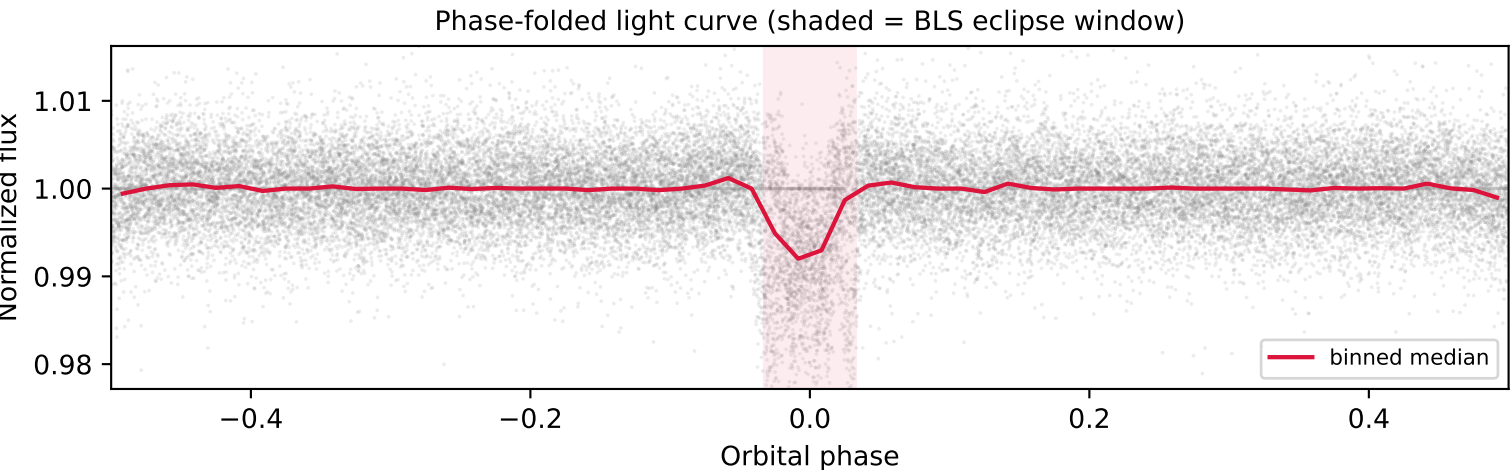
Sector 56 light-curve dip in this sector: 38.2 sigma t0 refined by 0.0075 phase

Difference-image peak SNR 87.24 (gate >= 5.0) Compactness 0.77 (gate >= 0.50)

Dimming centroid offset 0.18 px (4 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 2009125529

0.25 px away, dTmag +4.30 - farther from the dimming than the target



Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.066 px (dimming centroid 0.182 px from target vs 0.248 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.

TIC 303470667

Eclipsing binary candidate dossier - Star Vetter

MARGINAL

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	5.431910 d	VSX	no match
Epoch (t0)	1657.9950 BTJD	ASAS-SN	no match
Duration	0.1200 d	Gaia eclipsing-binary table	no match
Depth	2.220 % (22,198 ppm)	Gaia variable classification	no match
SNR	57.0	TESS EB catalog (Prsa+2022)	no match
INDEPENDENT CHECKS (f3i gates)		ExoFOP TOI list	no match
Eclipses observed	4 (gate >= 3)	Verdict:	NOVEL
Robust depth ratio	1.03 (gate >= 0.50)		
Robust depth sigma	38.6 (gate >= 7.0)		
Odd/even z-score	2.61 (consistent)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

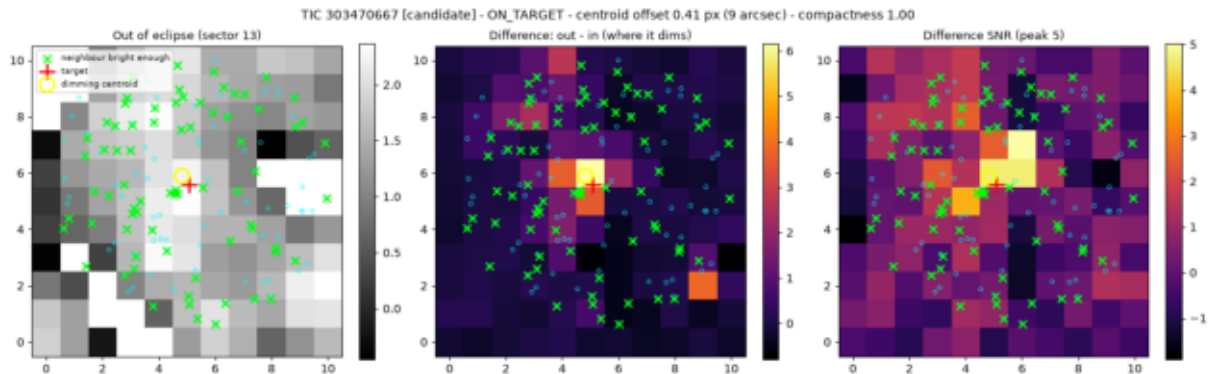
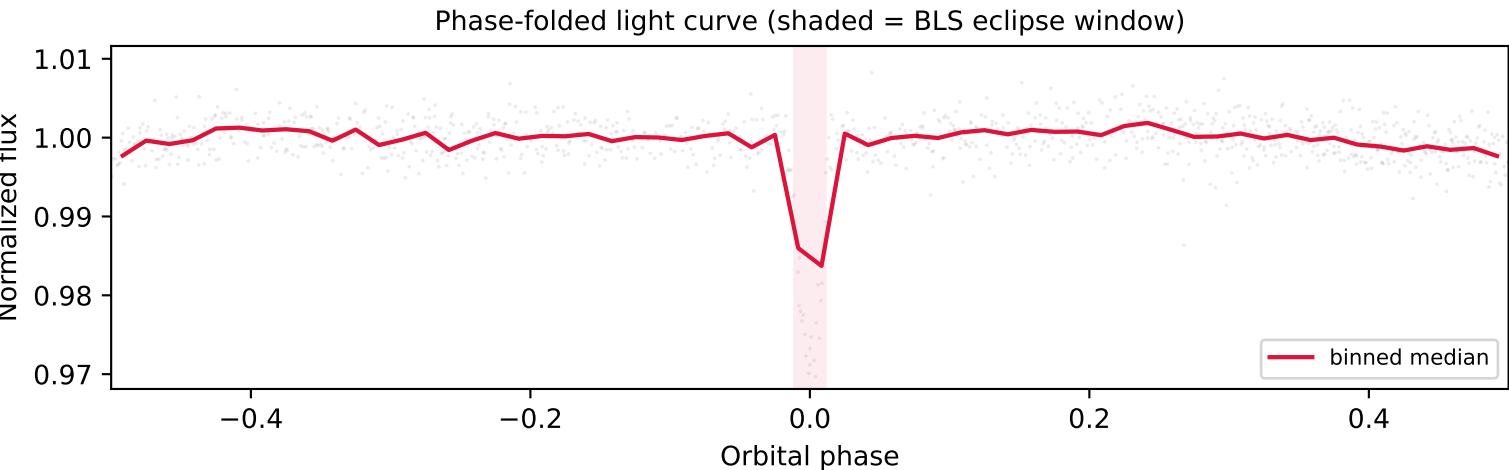
Sector 13 light-curve dip in this sector: 33.5 sigma t0 refined by 0.0025 phase

Difference-image peak SNR 5.02 (gate >= 5.0) Compactness 1.00 (gate >= 0.50)

Dimming centroid offset 0.41 px (9 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 1689765373

0.54 px away, dTmag +3.73 - farther from the dimming than the target



Caveat (MARGINAL): Pixel-check SNR is right at the gate (5.02 vs >= 5.0). A 150-trial false-alarm check on this star's own data found only a 3.3% chance of reaching this SNR at a random phase - the detection looks real, just faint. Weakest SNR margin of the confirmed set.

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	1.151883 d	VSX	no match
Epoch (t0)	1597.9084 BTJD	ASAS-SN	no match
Duration	0.0840 d	Gaia eclipsing-binary table	no match
Depth	1.255 % (12,546 ppm)	Gaia variable classification	no match
SNR	41.3	TESS EB catalog (Prsa+2022)	no match
INDEPENDENT CHECKS (f3i gates)		ExoFOP TOI list	no match
Eclipses observed	19 (gate >= 3)	Verdict: NOVEL	
Robust depth ratio	1.03 (gate >= 0.50)		
Robust depth sigma	28.6 (gate >= 7.0)		
Odd/even z-score	1.42 (consistent)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

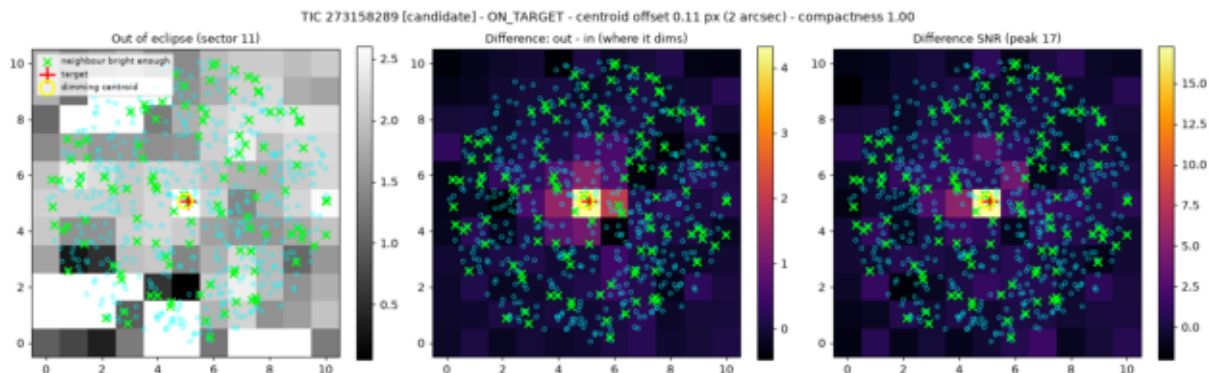
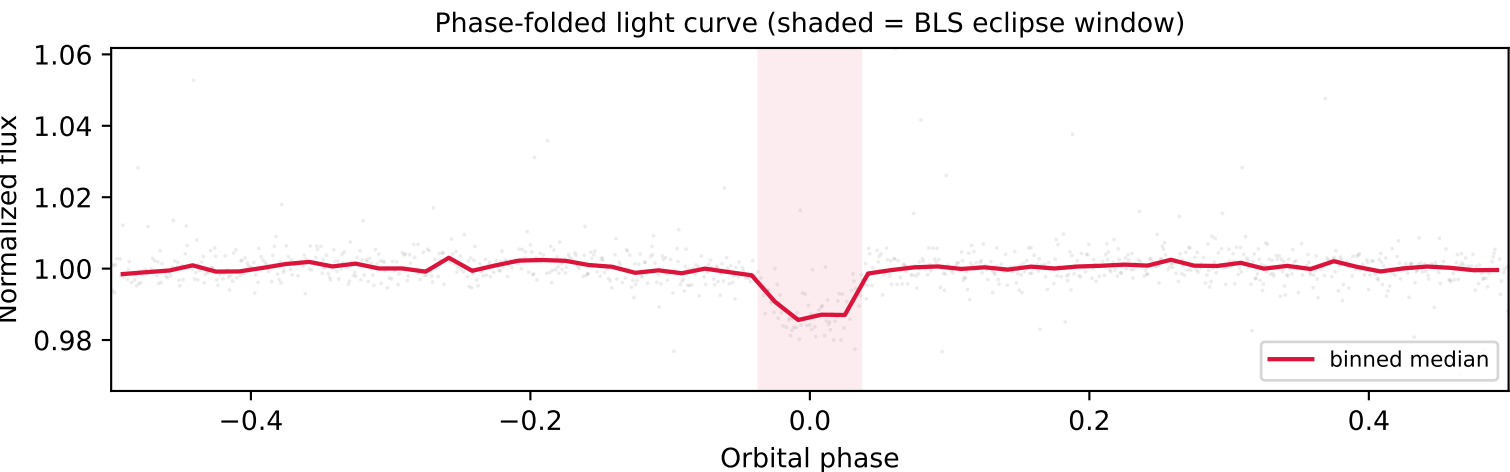
Sector 11 light-curve dip in this sector: 27.4 sigma t0 refined by 0.0050 phase

Difference-image peak SNR 17.24 (gate >= 5.0) Compactness 1.00 (gate >= 0.50)

Dimming centroid offset 0.11 px (2 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 514580937

0.12 px away, dTmag +0.71 - farther from the dimming than the target



Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.006 px (dimming centroid 0.113 px from target vs 0.119 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.

TIC 111149941

Eclipsing binary candidate dossier - Star Vetter

AMBIGUOUS PHOTOMETRY

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	5.240092 d	VSX	no match
Epoch (t0)	2695.2575 BTJD	ASAS-SN	no match
Duration	0.3500 d	Gaia eclipsing-binary table	no match
Depth	0.045 % (455 ppm)	Gaia variable classification	no match
SNR	10.2	TESS EB catalog (Prsa+2022)	no match
INDEPENDENT CHECKS (f3i gates)		ExoFOP TOI list	no match
Eclipses observed	3 (gate >= 3)	Verdict:	NOVEL
Robust depth ratio	1.02 (gate >= 0.50)		
Robust depth sigma	8.5 (gate >= 7.0)		
Odd/even z-score	2.29 (consistent)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

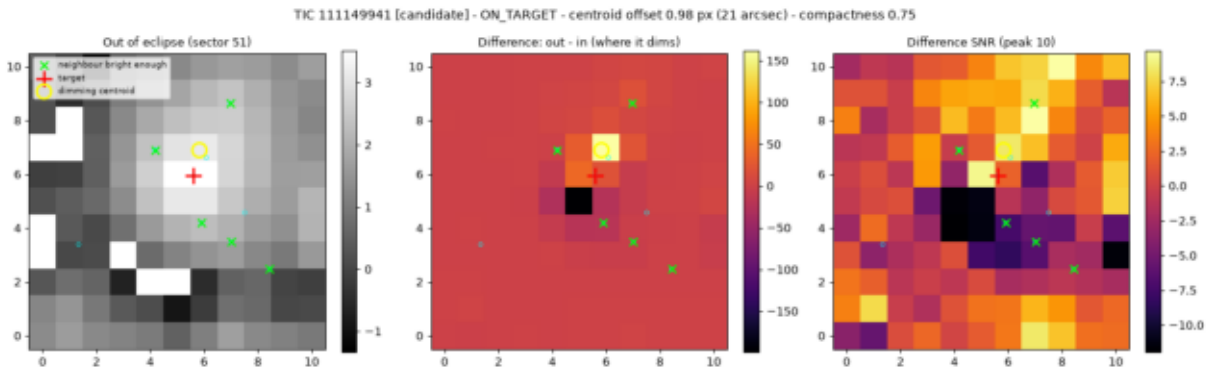
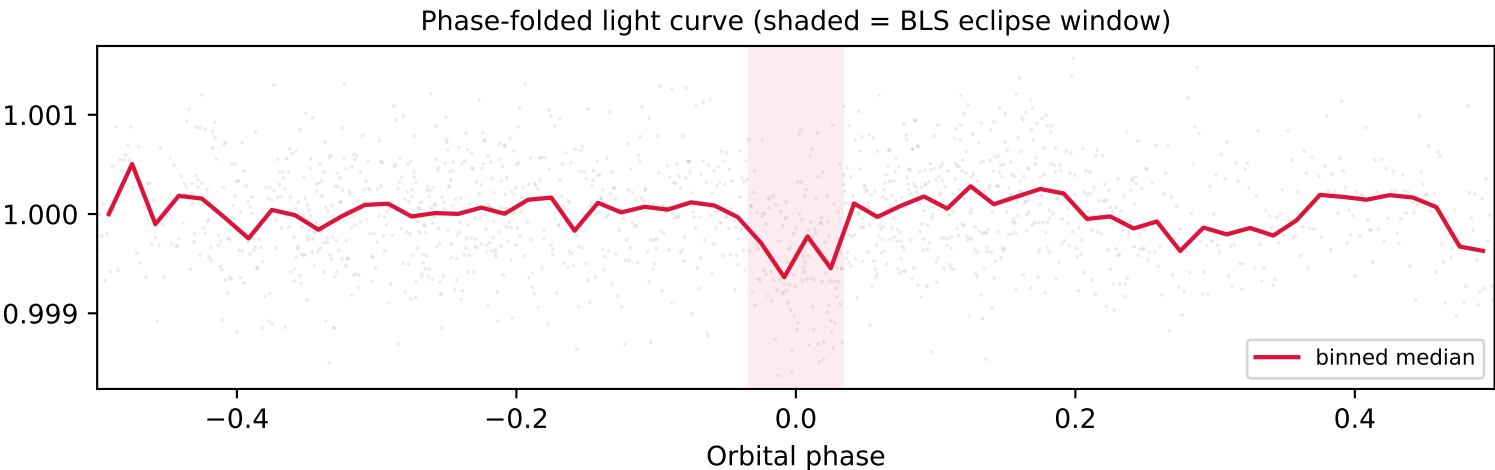
Sector 51 light-curve dip in this sector: 7.9 sigma t0 refined by 0.0100 phase

Difference-image peak SNR 9.73 (gate >= 5.0) Compactness 0.75 (gate >= 0.50)

Dimming centroid offset 0.98 px (21 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 1002359509

1.65 px away, dTmag +8.06 - farther from the dimming than the target



Caveat (AMBIGUOUS PHOTOMETRY): The difference-image detection is strong (peak SNR 9.7, compactness 0.75) but the star's own simple-aperture photometry shows no dip, and aperture/quality-mask sensitivity tests were inconsistent. Confirmed by pixel imaging; not independently confirmed by simple photometry.

TIC 251898081

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 27.308942 d (ALIASED, see below)
-> corrected period: 54.617884 d
Epoch (t0) 2876.1224 BTJD
Duration 0.5000 d
Depth 18.630 % (186,304 ppm)
SNR 5534.5

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

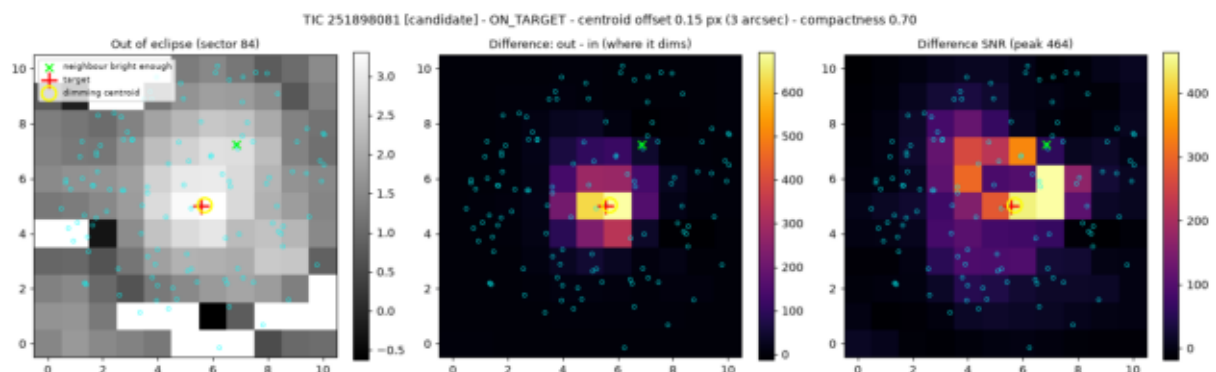
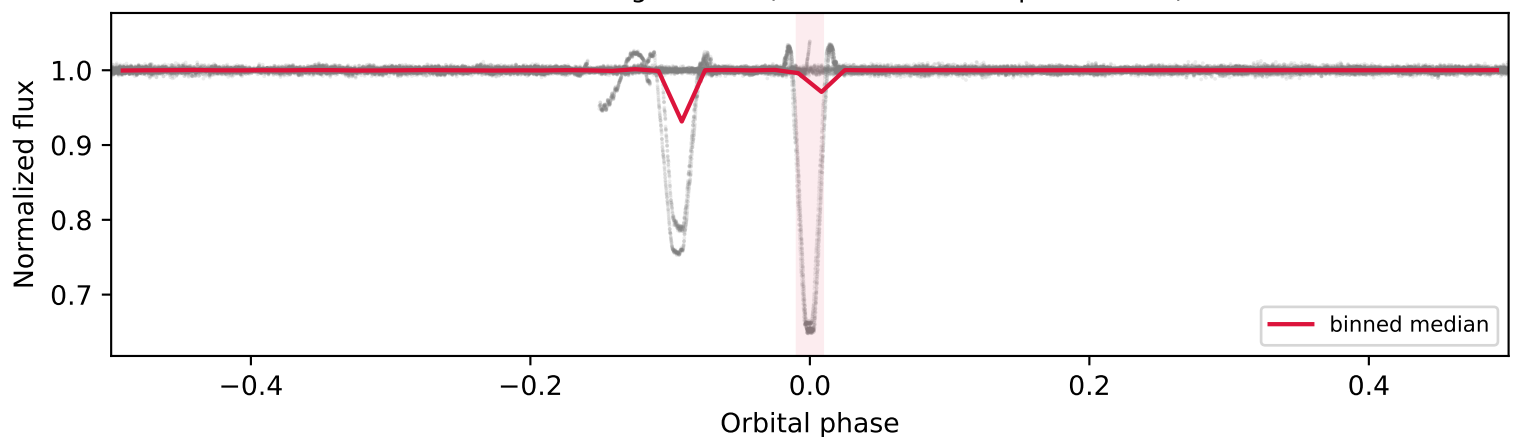
Eclipses observed 5 (gate ≥ 3)
Robust depth ratio 1.07 (gate ≥ 0.50)
Robust depth sigma 1819.7 (gate ≥ 7.0)
Odd/even z-score 198.85 (ALIAS FLAGGED)
Secondary eclipse detected, phase -0.09, 9.9 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 84 light-curve dip in this sector: 1459.8 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 464.16 (gate ≥ 5.0) Compactness 0.70 (gate ≥ 0.50)
Dimming centroid offset 0.15 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 251898060
2.48 px away, dTmag +1.74 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 198.85 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 54.617884 d, in any write-up.

TIC 467214801

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 0.760850 d (ALIASED, see below)
-> corrected period: 1.521701 d
Epoch (t0) 2825.3015 BTJD
Duration 0.0840 d
Depth 0.950 % (9,505 ppm)
SNR 2133.7

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

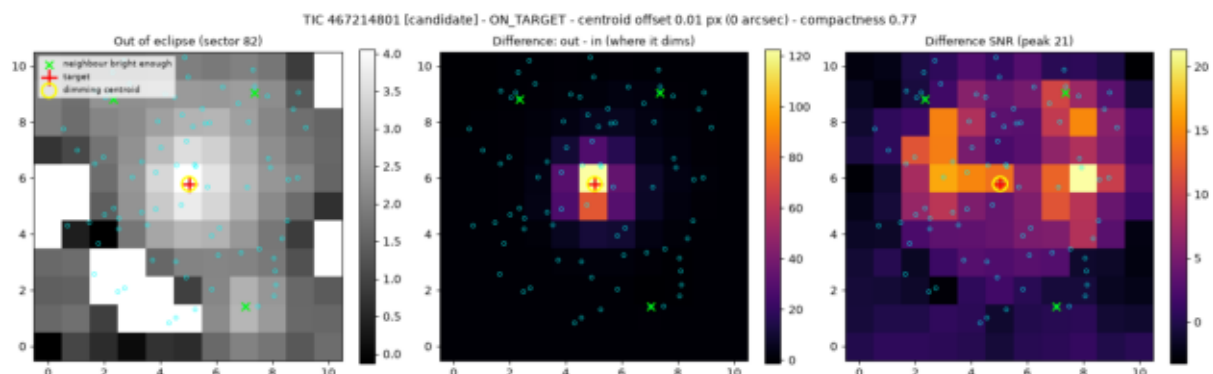
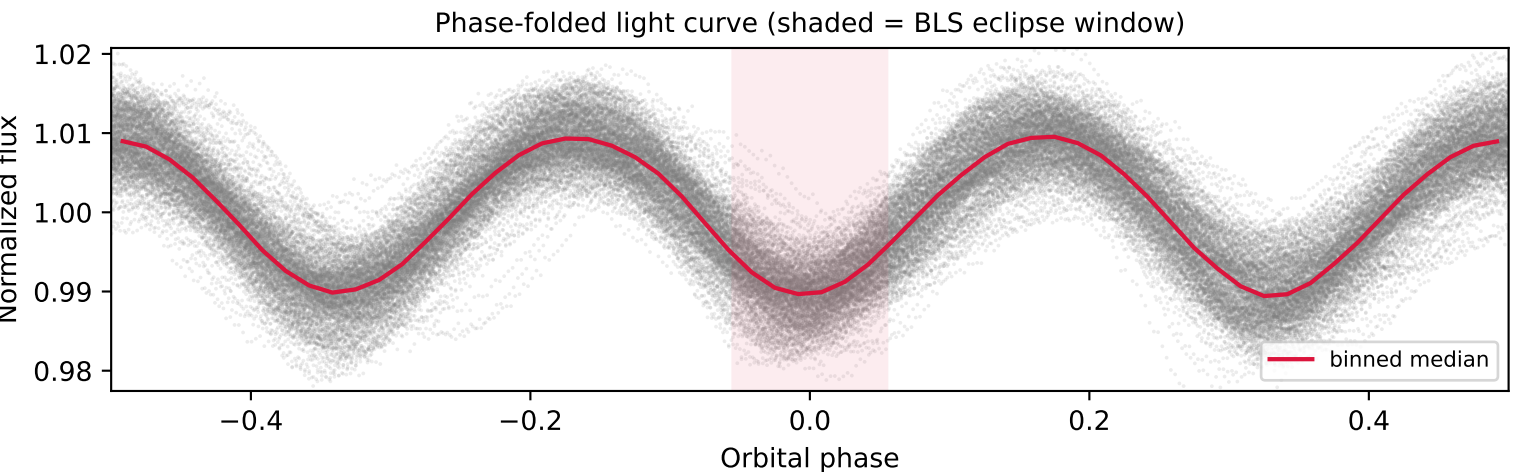
INDEPENDENT CHECKS (f3i gates)

Eclipses observed 329 (gate ≥ 3)
Robust depth ratio 1.11 (gate ≥ 0.50)
Robust depth sigma 104.6 (gate ≥ 7.0)
Odd/even z-score 5.40 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 82 light-curve dip in this sector: 25.9 sigma t0 refined by -0.0100 phase
Difference-image peak SNR 21.36 (gate ≥ 5.0) Compactness 0.77 (gate ≥ 0.50)
Dimming centroid offset 0.01 px (0 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 467214771
4.01 px away, dTmag +4.53 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 5.40 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 1.521701 d, in any write-up.

TIC 399770041

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 1.580189 d
Epoch (t0) 1569.9907 BTJD
Duration 0.0840 d
Depth 4.291 % (42,906 ppm)
SNR 488.6

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

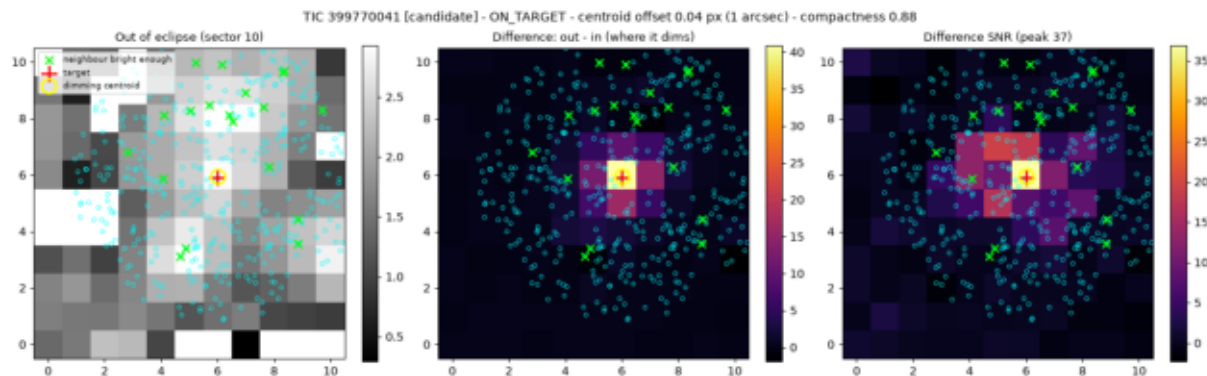
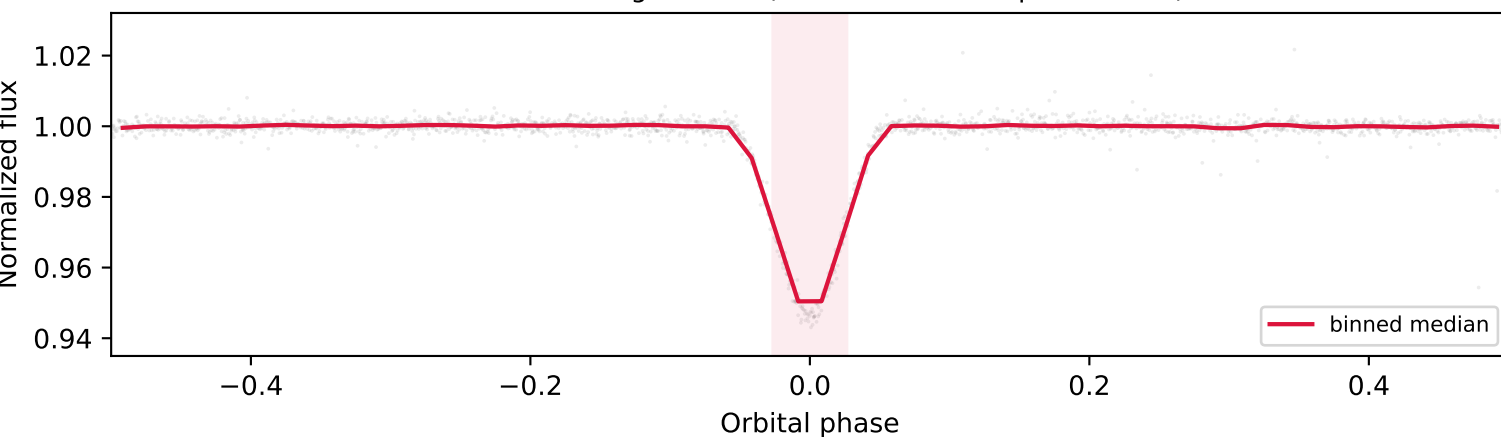
Eclipses observed 30 (gate ≥ 3)
Robust depth ratio 1.04 (gate ≥ 0.50)
Robust depth sigma 283.6 (gate ≥ 7.0)
Odd/even z-score 0.86 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 10 light-curve dip in this sector: 202.4 sigma t0 refined by -0.0075 phase
Difference-image peak SNR 36.92 (gate ≥ 5.0) Compactness 0.88 (gate ≥ 0.50)
Dimming centroid offset 0.04 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 399770066
1.83 px away, dTmag +2.10 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



TIC 349701853

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 30.342709 d
Epoch (t0) 2395.4286 BTJD
Duration 0.1200 d
Depth 28.445 % (284,452 ppm)
SNR 343.5

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

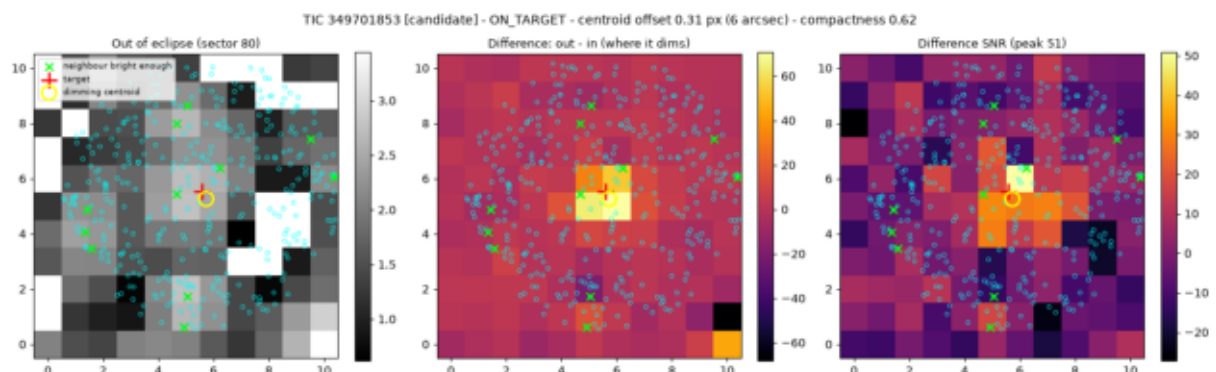
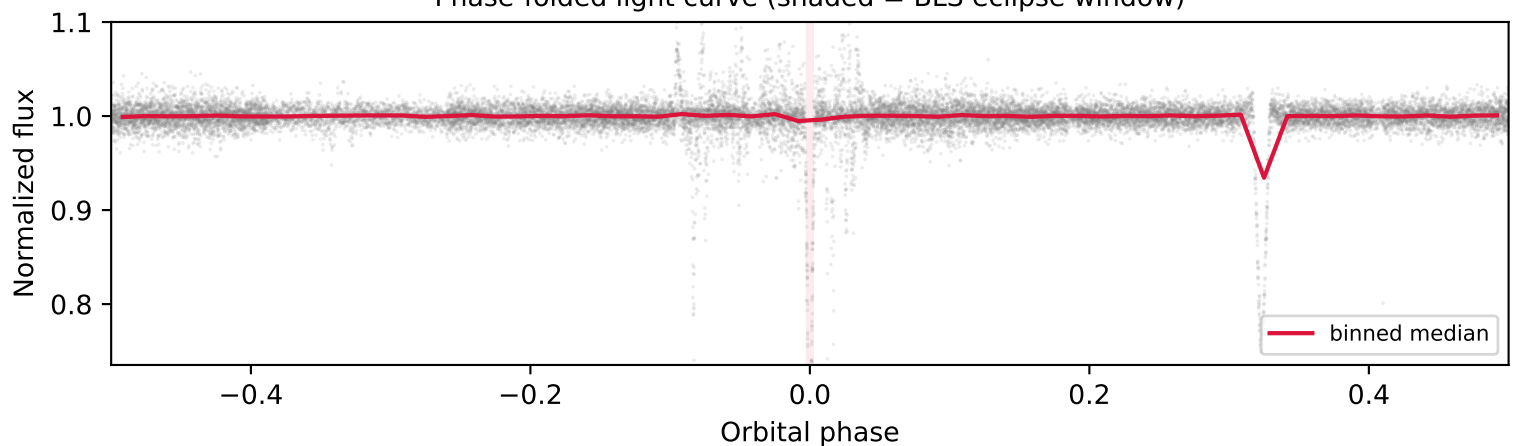
Eclipses observed 4 (gate ≥ 3)
Robust depth ratio 1.03 (gate ≥ 0.50)
Robust depth sigma 252.5 (gate ≥ 7.0)
Odd/even z-score 0.87 (consistent)
Secondary eclipse detected, phase 0.33, 10.0 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 80 light-curve dip in this sector: 149.7 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 51.05 (gate ≥ 5.0) Compactness 0.62 (gate ≥ 0.50)
Dimming centroid offset 0.31 px (6 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 349701859
1.07 px away, dTmag -0.58 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 233459038

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 25.338439 d (ALIASED, see below)
-> corrected period: 50.676878 d
Epoch (t0) 1728.5852 BTJD
Duration 0.1700 d
Depth 8.315 % (83,146 ppm)
SNR 236.6

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

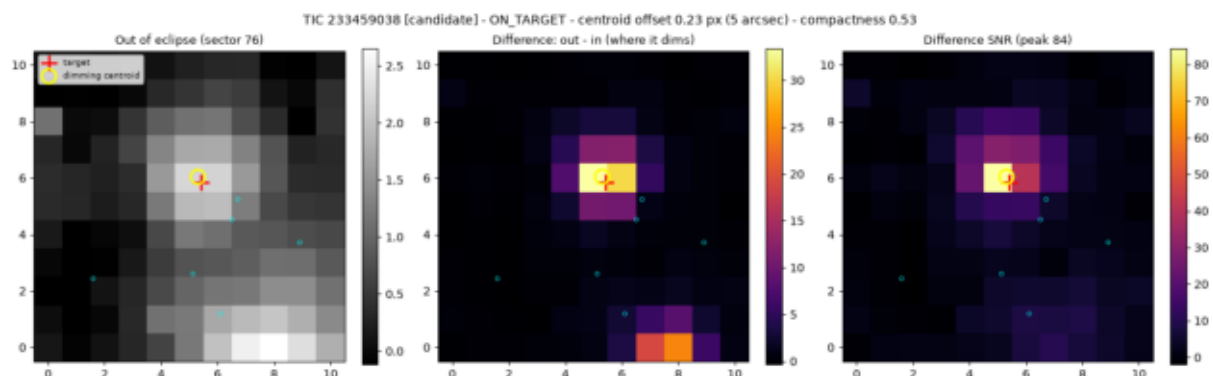
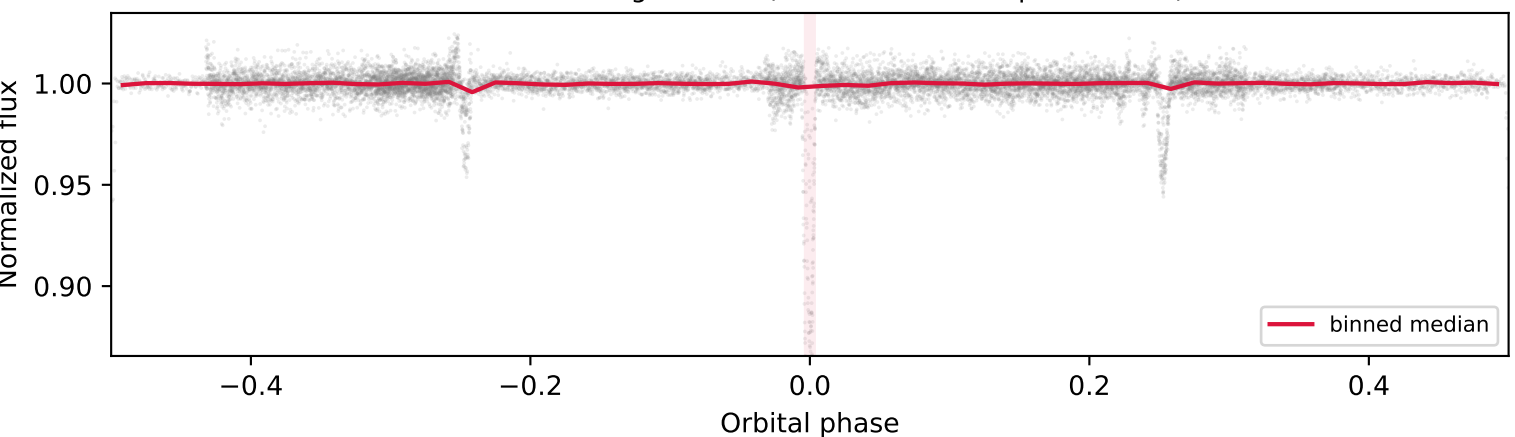
Eclipses observed 6 (gate ≥ 3)
Robust depth ratio 1.37 (gate ≥ 0.50)
Robust depth sigma 274.3 (gate ≥ 7.0)
Odd/even z-score 89.17 (ALIAS FLAGGED)
Secondary eclipse detected, phase 0.26, 8.3 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 76 light-curve dip in this sector: 169.1 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 84.46 (gate ≥ 5.0) Compactness 0.53 (gate ≥ 0.50)
Dimming centroid offset 0.23 px (5 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: none within the cutout

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 89.17 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 50.676878 d, in any write-up.

TIC 281310348

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 1.770214 d
Epoch (t0) 1469.3509 BTJD
Duration 0.1200 d
Depth 7.273 % (72,730 ppm)
SNR 179.7

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 11 (gate ≥ 3)
Robust depth ratio 0.98 (gate ≥ 0.50)
Robust depth sigma 114.9 (gate ≥ 7.0)
Odd/even z-score 0.16 (consistent)
Secondary eclipse not detected

CATALOG CROSS-MATCH (novelty)

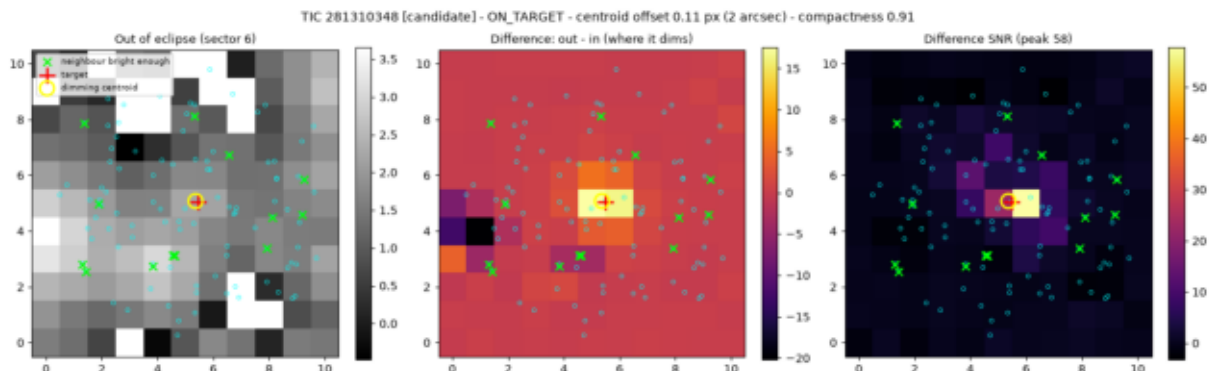
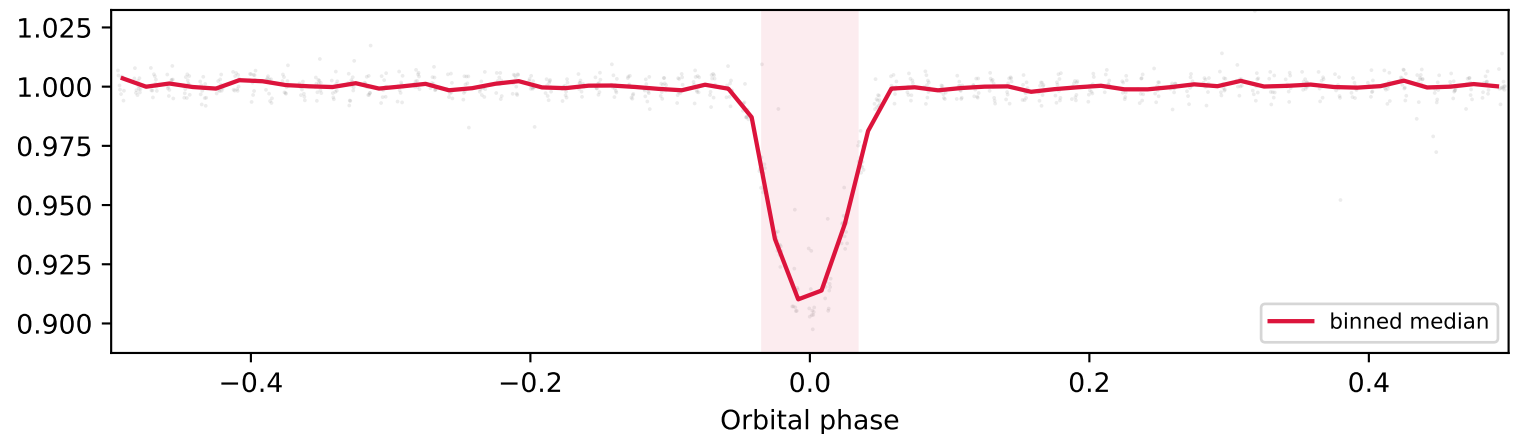
VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 6 light-curve dip in this sector: 133.0 sigma t0 refined by 0.0100 phase
Difference-image peak SNR 57.87 (gate ≥ 5.0) Compactness 0.91 (gate ≥ 0.50)
Dimming centroid offset 0.11 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 281310326
2.04 px away, dTmag +1.77 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 318887646

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 14.281664 d
Epoch (t0) 1418.6279 BTJD
Duration 0.1200 d
Depth 3.605 % (36,049 ppm)
SNR 149.9

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 3 (gate ≥ 3)
Robust depth ratio 1.20 (gate ≥ 0.50)
Robust depth sigma 143.4 (gate ≥ 7.0)
Odd/even z-score 1.52 (consistent)
Secondary eclipse not detected

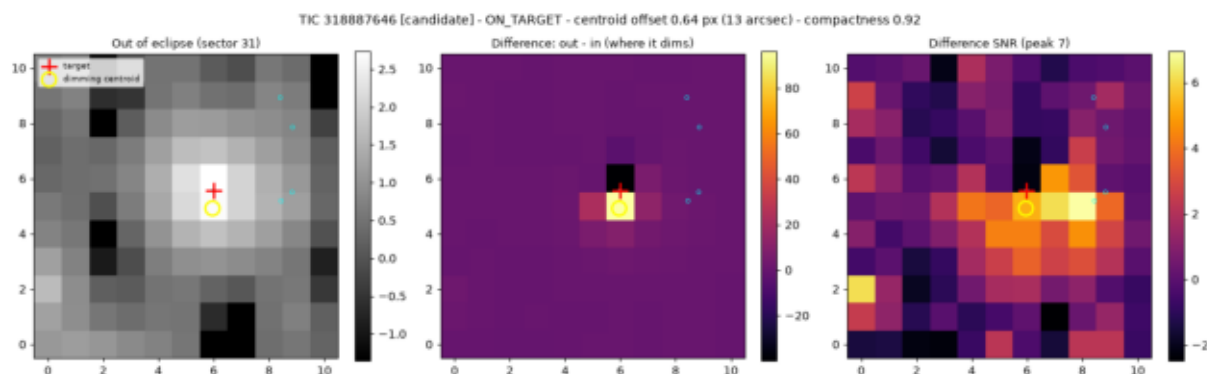
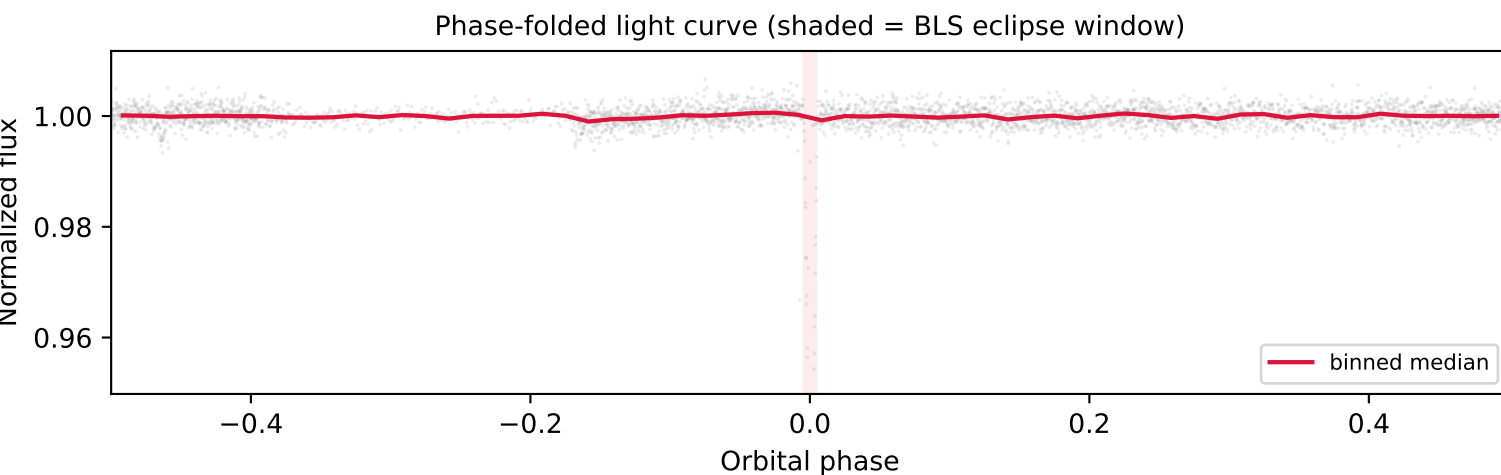
CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 31 light-curve dip in this sector: 124.6 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 6.94 (gate ≥ 5.0) Compactness 0.92 (gate ≥ 0.50)
Dimming centroid offset 0.64 px (13 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: none within the cutout



No caveats - clean detection on every check.

TIC 444465823

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 18.513330 d (ALIASED, see below)
-> corrected period: 37.026660 d
Epoch (t0) 1768.5512 BTJD
Duration 0.1200 d
Depth 3.476 % (34,755 ppm)
SNR 146.6

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

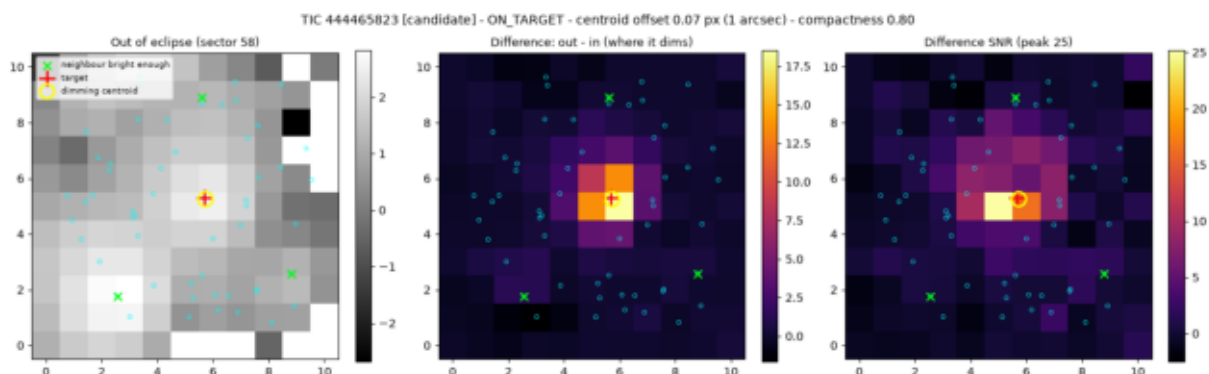
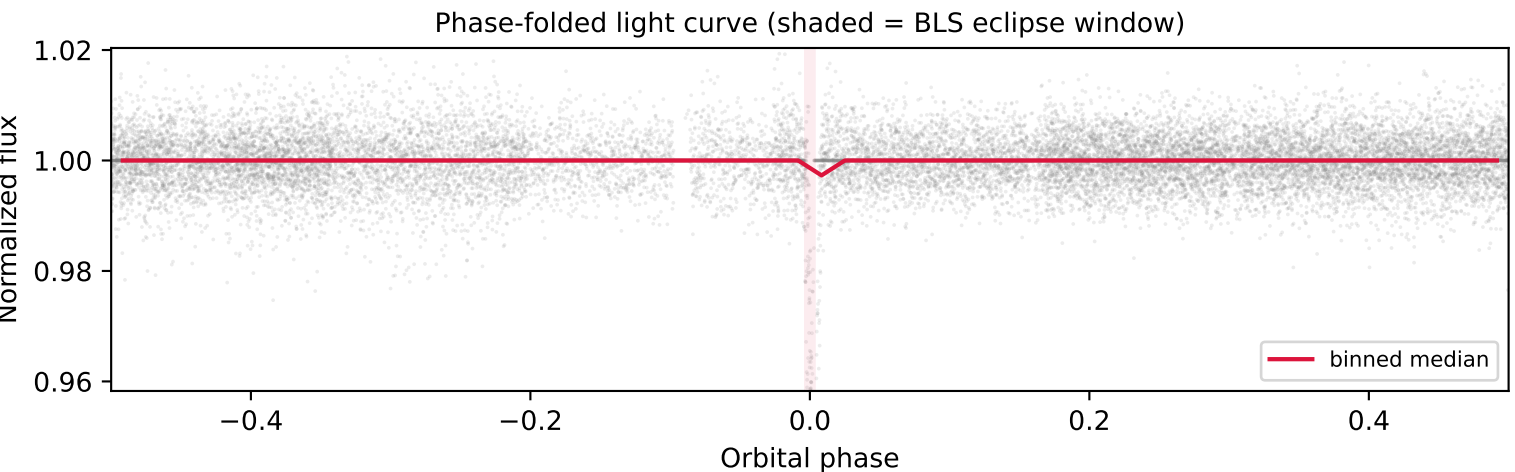
INDEPENDENT CHECKS (f3i gates)

Eclipses observed 8 (gate ≥ 3)
Robust depth ratio 1.01 (gate ≥ 0.50)
Robust depth sigma 95.7 (gate ≥ 7.0)
Odd/even z-score 14.96 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 58 light-curve dip in this sector: 61.8 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 25.18 (gate ≥ 5.0) Compactness 0.80 (gate ≥ 0.50)
Dimming centroid offset 0.07 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 444465799
3.65 px away, dTmag +2.93 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 14.96 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 37.026660 d, in any write-up.

TIC 380943251

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 0.720935 d
Epoch (t0) 1386.4490 BTJD
Duration 0.0580 d
Depth 7.667 % (76,670 ppm)
SNR 101.0

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

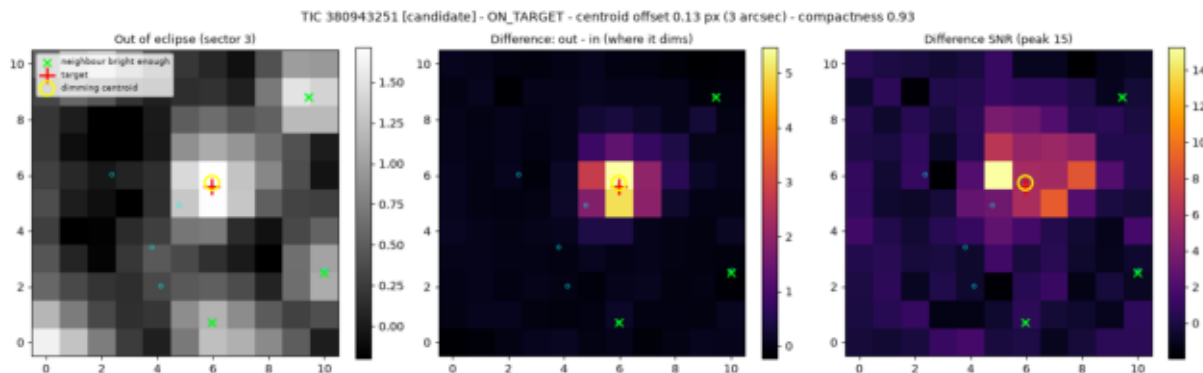
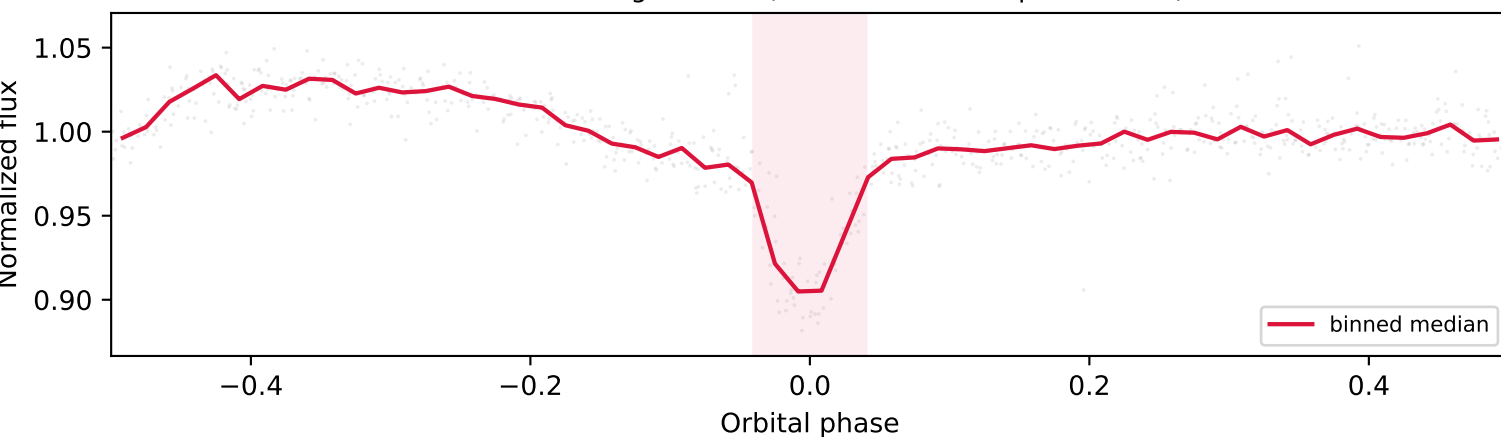
Eclipses observed 14 (gate ≥ 3)
Robust depth ratio 1.03 (gate ≥ 0.50)
Robust depth sigma 27.8 (gate ≥ 7.0)
Odd/even z-score 1.89 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 3 light-curve dip in this sector: 25.9 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 15.26 (gate ≥ 5.0) Compactness 0.93 (gate ≥ 0.50)
Dimming centroid offset 0.13 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 380941073
4.65 px away, dTmag +0.59 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



TIC 66716482

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 8.028259 d
Epoch (t0) 2561.2964 BTJD
Duration 0.1200 d
Depth 3.329 % (33,287 ppm)
SNR 64.9

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

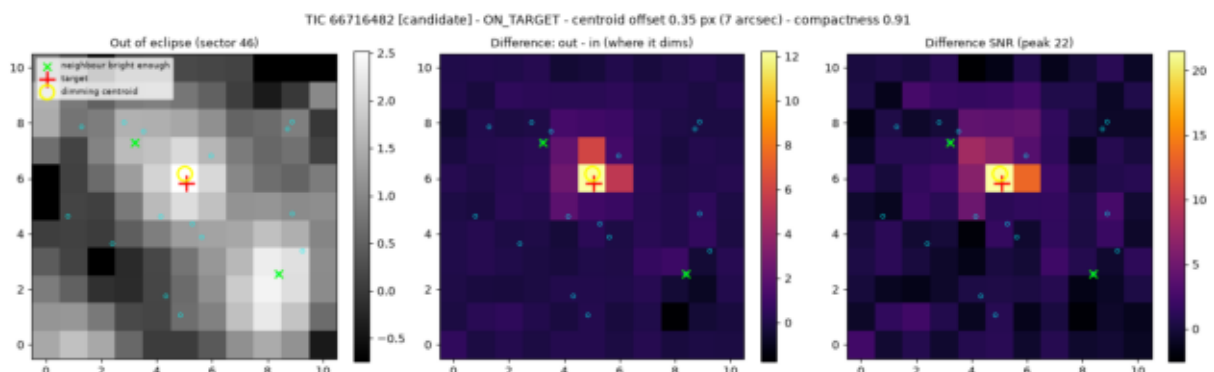
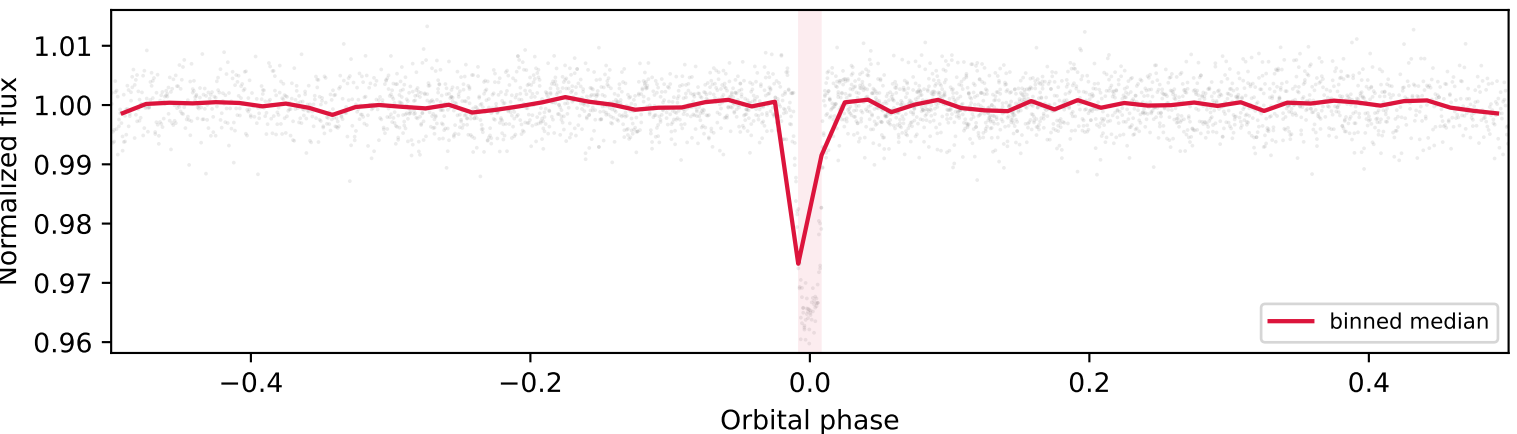
Eclipses observed 3 (gate ≥ 3)
Robust depth ratio 1.03 (gate ≥ 0.50)
Robust depth sigma 56.0 (gate ≥ 7.0)
Odd/even z-score 0.64 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 46 light-curve dip in this sector: 44.5 sigma t0 refined by -0.0025 phase
Difference-image peak SNR 21.51 (gate ≥ 5.0) Compactness 0.91 (gate ≥ 0.50)
Dimming centroid offset 0.35 px (7 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 66715933
2.13 px away, dTmag +2.34 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 309656476

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 28.920445 d
Epoch (t0) 2997.6132 BTJD
Duration 0.1700 d
Depth 0.050 % (495 ppm)
SNR 24.6

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

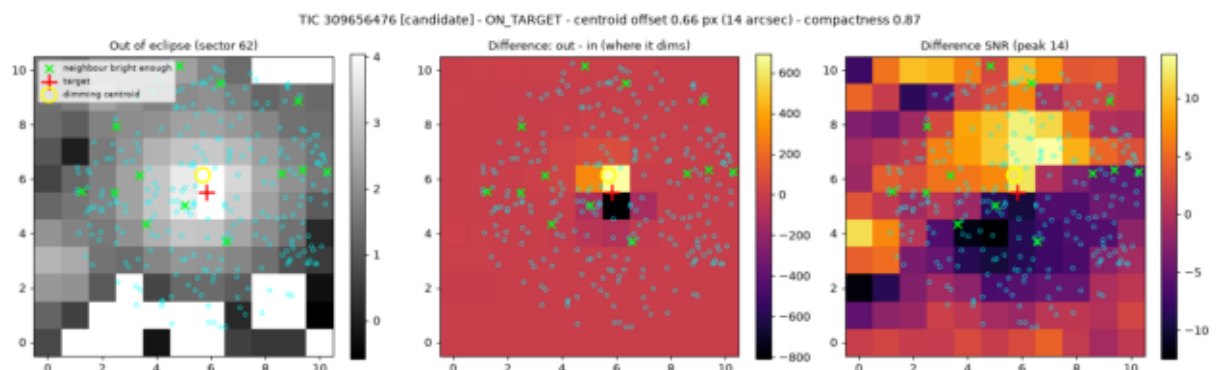
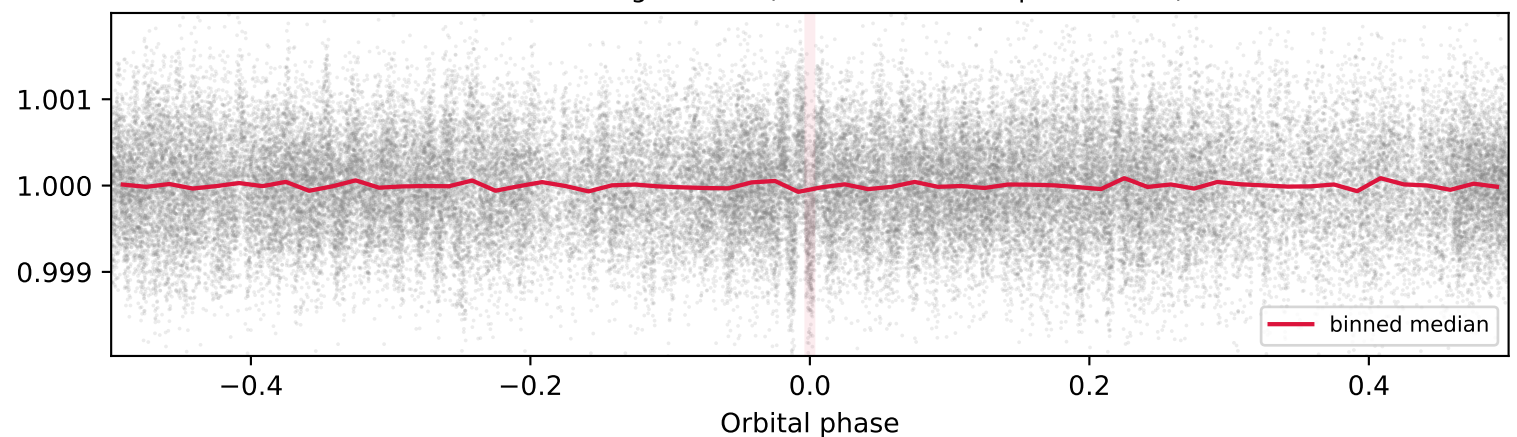
Eclipses observed 6 (gate ≥ 3)
Robust depth ratio 0.98 (gate ≥ 0.50)
Robust depth sigma 14.1 (gate ≥ 7.0)
Odd/even z-score 1.97 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 62 light-curve dip in this sector: 8.5 sigma t0 refined by 0.3975 phase
Difference-image peak SNR 13.80 (gate ≥ 5.0) Compactness 0.87 (gate ≥ 0.50)
Dimming centroid offset 0.66 px (14 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 309656481
1.29 px away, dTmag +5.82 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 229494833

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 15.383868 d (ALIASED, see below)
-> corrected period: 30.767736 d
Epoch (t0) 2829.9134 BTJD
Duration 0.1200 d
Depth 0.037 % (372 ppm)
SNR 14.9

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

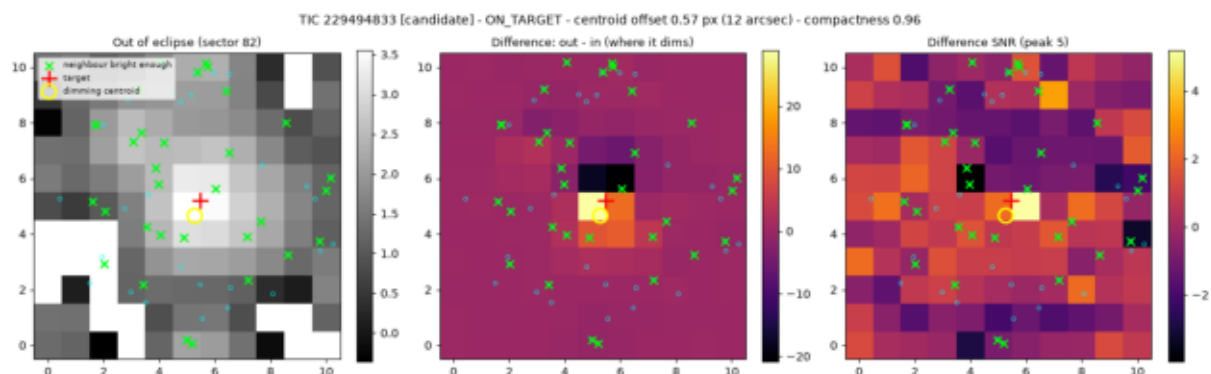
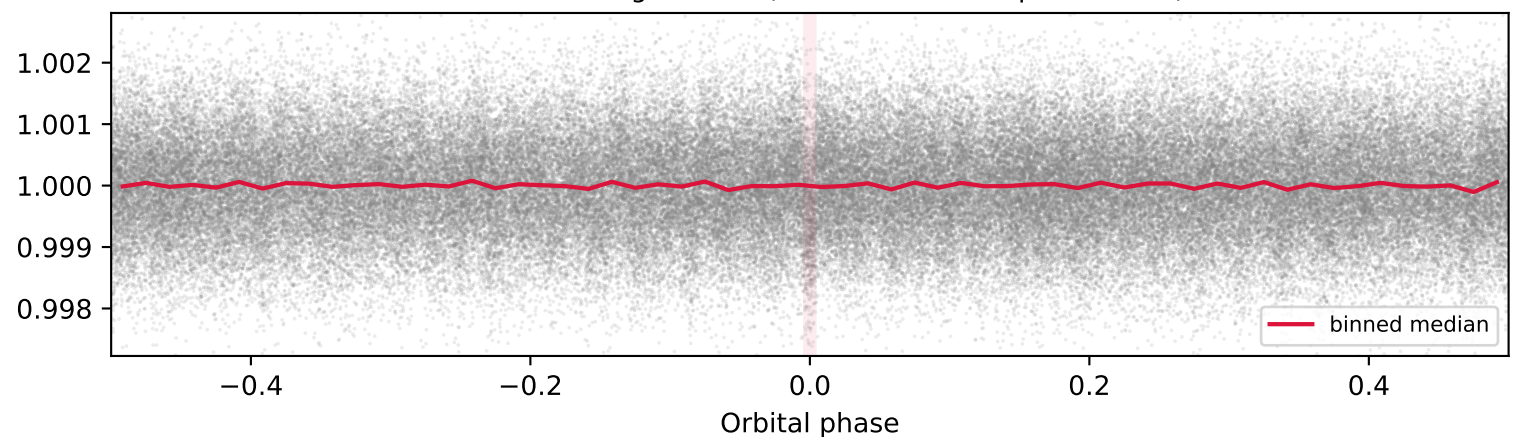
Eclipses observed 19 (gate ≥ 3)
Robust depth ratio 1.00 (gate ≥ 0.50)
Robust depth sigma 11.2 (gate ≥ 7.0)
Odd/even z-score 4.32 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 82 light-curve dip in this sector: 6.8 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 5.19 (gate ≥ 5.0) Compactness 0.96 (gate ≥ 0.50)
Dimming centroid offset 0.57 px (12 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 229494826
0.88 px away, dTmag +4.84 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 4.32 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 30.767736 d, in any write-up.

TIC 44533527

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 14.595189 d
Epoch (t0) 3463.8871 BTJD
Duration 0.1200 d
Depth 0.101 % (1,013 ppm)
SNR 12.0

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

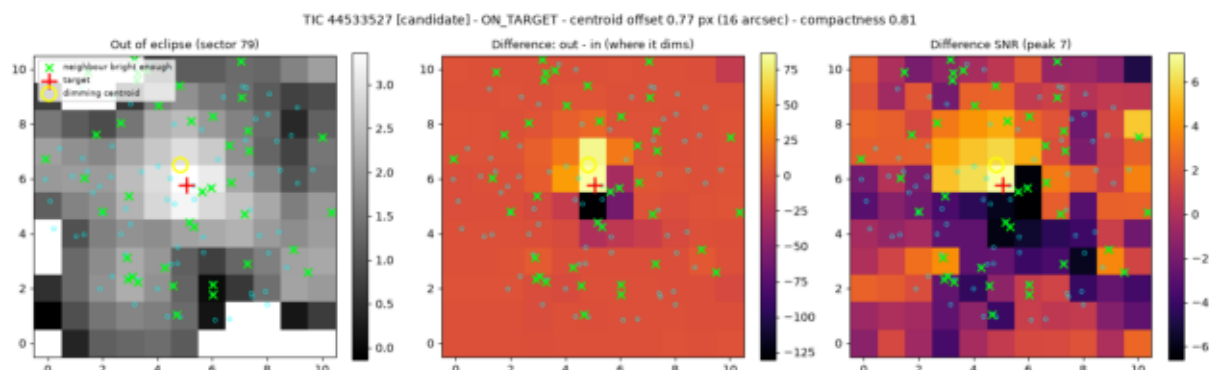
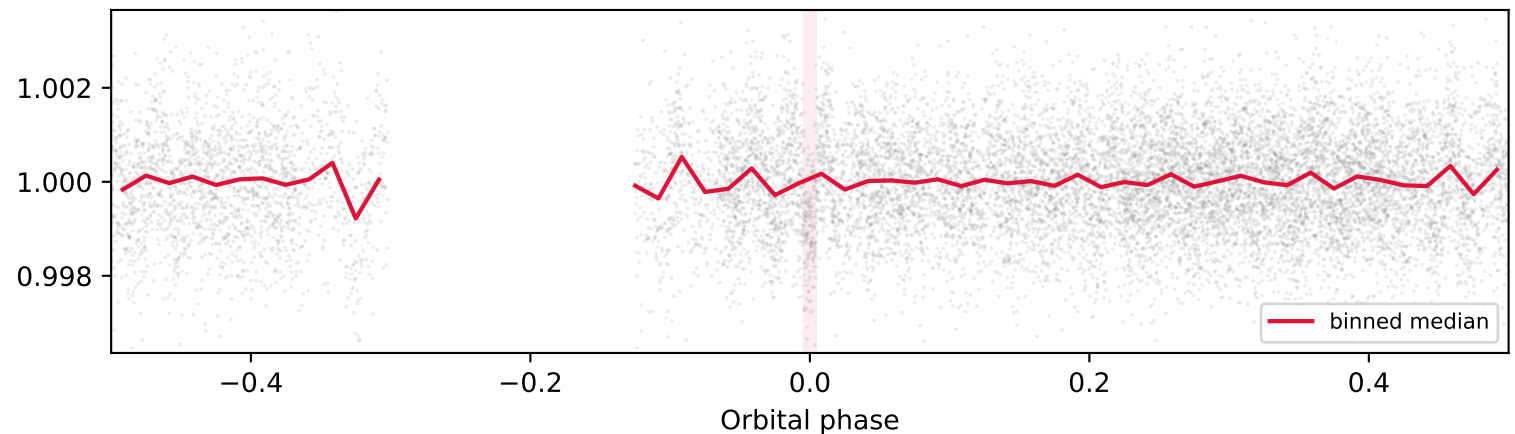
Eclipses observed 3 (gate ≥ 3)
Robust depth ratio 0.94 (gate ≥ 0.50)
Robust depth sigma 8.6 (gate ≥ 7.0)
Odd/even z-score 1.18 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 79 light-curve dip in this sector: 6.8 sigma t0 refined by -0.3250 phase
Difference-image peak SNR 7.33 (gate ≥ 5.0) Compactness 0.81 (gate ≥ 0.50)
Dimming centroid offset 0.77 px (16 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 44533532
1.26 px away, dTmag +4.17 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.